

NovaCart Grounded Metrics Agent

A retrieval-augmented n8n workflow that answers NovaCart leadership's metrics questions strictly from internal documents — and refuses rather than guesses when it can't find the answer.

Problem

NovaCart's weekly metrics (revenue, conversion, churn, SLA) are scattered across Drive documents. Manually compiling them is slow, and generic LLM tools hallucinate numbers when they lack real access to the data.

Approach & Key Decisions

- Two linked n8n pipelines share one Pinecone index: an ingestion pipeline (Drive → chunk → embed → upsert) and a retrieval pipeline (chat → forced tool call → grounded answer).
- Agent system prompt forces exactly one retrieval-tool call before any substantive answer, and an exact refusal string when retrieval comes back empty or off-topic.
- Trade-off: this sacrifices conversational flexibility for auditability — a predictable "I don't have that" beats a plausible-sounding wrong number in a leadership reporting tool.
- Cohere reranking sits between Pinecone and the agent to improve the quality of the top-K=5 retrieved chunks before they reach the model.

What It Does

- Syncs a Pinecone vector index from a Google Drive folder on a schedule or via manual form upload.
- Serves a public, custom-branded, streaming chat interface backed by GPT-4.1.
- Retrieves top-5 chunks per question, reranks with Cohere, and answers only from that retrieved text.
- Falls back to web search only for explicit external/competitor comparison questions.

Results & Learnings

A full real execution (see docs/Screenshots.pdf) shows the ingestion pipeline populating Pinecone and three separate chat prompts returning grounded answers — e.g. correctly breaking down NovaPhone X1's USD 1.62B revenue share vs. lagging product lines from the indexed annual report and sales data. Forcing one tool call plus an exact refusal string proved a cheap, effective way to make "I don't know" predictable. Next steps: widen ingestion beyond Drive (Notion/Confluence), add authentication before real rollout, and surface citation metadata in the final answer.

Tech Stack

Layer	Technology
Orchestration	n8n workflow (two linked pipelines)
Source storage	Google Drive (OAuth2)
Chunking	Recursive Character Text Splitter (500 / 50 overlap)
Embeddings	OpenAI text-embedding-3-large (1024-dim)
Vector DB	Pinecone (index: vidyanovacart, namespace: PROD, top-K 5)
Reranker	Cohere Reranker
Agent / LLM	OpenAI gpt-4.1 (LangChain Agent node)
Memory	n8n Buffer Window Memory

How to Run: import codebase/02-RAG-NovacartRAGIngestionANDRetrievalPipelines.json into n8n, attach Google Drive / OpenAI / Pinecone / Cohere credentials, create a 1024-dim Pinecone index, run ingestion once, then activate and open the Chat Trigger URL.